1N5819

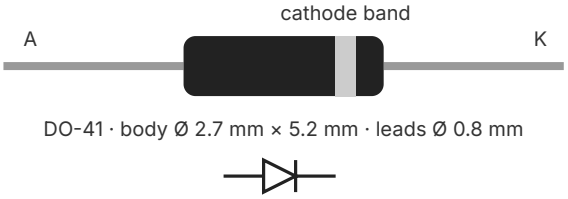
Schottky barrier rectifier · 1 A · 40 V

Datasheet summary compiled from typical manufacturer data (T_A = 25 °C unless noted) – for teaching purposes. Not an official manufacturer datasheet; values of real parts may differ slightly.

Features

- Metal–semiconductor (Schottky) junction
- Low forward voltage
- For low-voltage rectification, reverse-polarity and battery protection
- Note: higher reverse current than PN diodes, strongly temperature dependent

Package



DO-41 · body Ø 2.7 mm × 5.2 mm · leads Ø 0.8 mm

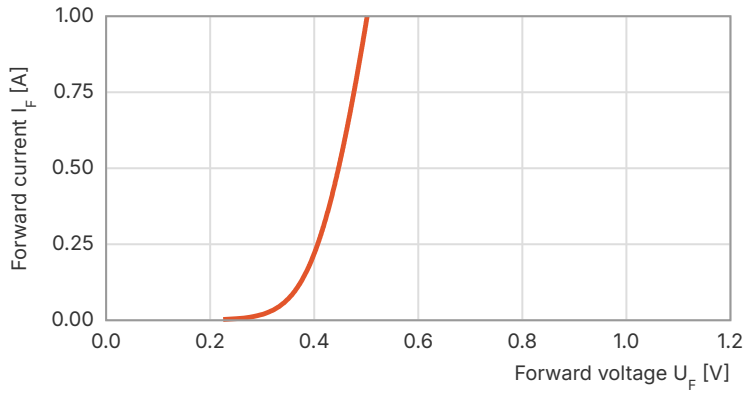
Absolute maximum ratings

Parameter	Symbol	Value	Unit
Repetitive peak reverse voltage	U_{RRM}	40	V
RMS reverse voltage	$U_{R(RMS)}$	28	V
Average forward current	$I_{F(AV)}$	1.0	A
Peak forward surge current, 8.3 ms half sine	I_{FSM}	25	A
Junction temperature	T_j	−65 … +125	°C

Electrical characteristics

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Forward voltage	U_F	I_F = 0.5 A	–	0.45	–	V
Forward voltage	U_F	I_F = 1.0 A	–	0.50	0.60	V
Forward voltage	U_F	I_F = 3.0 A	–	–	0.90	V
Reverse current	I_R	$U_R = U_{RRM}$, T_j = 25 °C	–	–	1	mA
Reverse current	I_R	$U_R = U_{RRM}$, T_j = 100 °C	–	–	10	mA

Typical forward characteristic (T_j = 25 °C)



Typical curve at 25 °C. The forward voltage rises with the current; at higher junction temperature it becomes slightly lower.